Computer organization and architecture

UNIT-I

2 MARKS

1. What are the functional units?
2. what is meant by input unit?
3. What is meant by memory unit?
4. What are the operations in ALU?
5. What is meant by output unit?
6. What are the operations in control unit?
7. Define byte addressability?
8. What is meant by multiprocessor?
9. What is meant by multicomputers?
10. What is meant by Big-Endian?
11. What is meant by little endian?
12. How to align the word?
13. What are the Ways to indicate the length of the string?
14. What are the memory operations?
15. What are the instruction sequencing?
16. What is meant by Register Transfer Notation?
17. What is meant by Assembly language notation?
18. What are the basic instruction types?
19. How to evaluate(A+B) *(C+D) using RISC?
20. What is meant by straight-line sequencing?
21. What are the phases to execute the instruction?
22. What are the flags in condition codes?
23. What is meant by addressing modes?
24. What is meant by index mode and relative mode?
25. What is meant by subroutine lingage and link register?
26. What is meant by subroutine nesting?

11 MARKS

1. Explain the various types of addressing mode with example.(Nov 2015,April 2015)
2. Explain the shift and rotate operations with suitable example.(Nov 2015,2014)
3. With a neat diagram explain the functional units of a computer .89.(Nov 2014)
4. What are special registers in a typical computer? Explain their purpose with example.(April 2015)
5. Describe the organization of stack(4 Marks April 2015)
6. Explain with a diagram the design of a fast multiplier using carry save adder circuit.(April 2015)
7. Write in detail about functional units of computer.

8. Write in detail about basic Input\output operations.
9. Explain in detail about multicomputers and microprocessors.
10. Write in detail about instruction sequencing.

UNIT-II

2 MARKS

1. How to access memory in IA-32 Pentium?
2. what are the registers used in IA-32 Memory?
3. What are the addressing modes in IA-32?
4. What is meant by one byte instructions?
5. What is meant by immediate mode encoding?
6. What are the IA-32 Assembly language?
7. What is meant by conditional jumps?
8. What is meant by unconditional jumps?
9. What are the logic operations?
10. What are the shift and rotate operations?
11. What are the shift instructions?
12. Explain the digit packing using examples?
13. Explain memory – mapped I/O?
14. Explain Isolated I/O?
15. What are the four instructions to push and pop?
16. What are the multiply and divide instructions?
17. What are the Multimedia extension (MMX) instructions?
18. What is meant by vector instructions?
19. What is meant vector dot product program?
20. What is meant by Byte- sorting program?
21. What is meant by linked-list insertion and deletion subroutines?

11 MARKS

1. Explain the architecture of IA-32 with s
   suitable block diagram.(Nov 2015)
2. Explain the operations IA-32 with example.(Nov 2015,2014)
3. Explain the machine instruction format of IA-32.(Nov 2014,April 2015)
4. Explain in detail about the various addressing modes available in IA-32 Pentium machine with example.(April 2015)
5. Write an Assemble code to check whether the data A is positive or negative number.(April 2015)
6. Write an assembly code to check whether the data X is odd parity or even parity data.(April 2015)
7. Write in detail about logic and shift instructions of IA-32.
8. Discuss in detail about subroutines.
9. Briefly explain about (i) Registers (ii) Addressing

10. Explain in detail about subroutines of IA-32.

UNIT-III

2 MARKS

1. What is meant by Interrupts?
2. What is meant by Direct Memory Access?
3. What is meant by interrupt hardware?
4. How to enable and disable interrupts?
5. What are the issues will rise when handle with multiple device?
6. What is meant by vectored interrupts?
7. What is meant interrupt nesting?
8. What is meant by simultaneous requests?
9. How to control device requests?
10. What are the terms used in exceptions?
11. What are the registers in DMA interface?
12. What is meant by bus arbitration?
13. What are the two approaches in arbitrations?
14. What is meant by scheme in bus?
15. What is meant by synchronous bus?
16. What is meant by Asynchronous bus?
17. How to interface circuits?
18. What are the functions of the I/O interface?
19. What are the serial interface circuit involves in?
20. What are the standard I/O interfaces?
21. What are the limitations of the Port?
22. What are the characteristics of Device?
23. How to plug – and – play?

11 MARKS

1. Explain the interrupt enabling and disabling mechanism.((Nov 2015)
2. Explain the process of DMA with suitable diagram.(Nov 2015,2014,April 2015)
3. What is an exception? Explain in detail about the types of exceptions.(Nov 2014)
4. Explain the protocols of USB with a neat diagram.(April 2014)
5. Describe the hardware mechanism for handling multiple interrupt requests.(April 2015)
6. What are handshaking signals? Explain the handshake control of data transfer during input and output operation.(April 2015)
7. Explain in detail about Pentium interrupt structure.(April 2015)
8. What are the different methods of I/O transfer and explain how is data transfer done using DMA.
9. Explain about buses in detail.

10. Write short notes on Interrupt and Interrupt Hardware. Write a program that reads a line from the keyboard and echoes it back to display.

UNIT-IV

2 MARKS

1. What are the techniques to increase the size and speed of the memory?
2. What is SRAM cell?
3. What is meant by dynamic RAMs?
4. What is meant by fast page mode?
5. What is known as memory controller and what are their parts?
6. What is meant by cache memory?
7. What is meant by locality of reference and what are their types?
8. What is meant by cache hit?
9. What is known as cache miss?
10. What are all the mapping functions?
11. What is meant by write buffer?
12. What is meant by prefetching?
13. What is meant by lockup-Free cache?
14. What is meant virtual memories?
15. What is meant by page table, page frame and page table base register?
16. What are the states in processor?
17. What are the RAID Disk Arrays?

11 Marks

1. Explain the memory hierarchy with respect to speed, size and cost.(Nov 2015)
2. Explain the concept of virtual memories.(Nov 2015,2014)
3. Discuss the different types of cache mapping techniques that are employed. Point out heir relative advantages and disadvantages.(Nov 2015,2014)
4. Discuss in detail about semiconductor RAM memory.
5. Explain the organization and accessing of data on a disk.(April 2015)
6. Explain how the virtual address is converted into real address in a paged virtual memory system.(April 2015)
7. Discuss the different types of ROM.((Nov 2014)
8. Explain in detail about secondary storage.
9. Discuss in detail about memory management requirements.
10. Describe the virtual memory address translation technique.(April 2013)

UNIT-V

2 MARKS

1. What are the fundamental concepts of basic processing units?

Sketch the basic instruction cycle?

2. What are the operations to be performed for execution of various instructions?
3. Explain the operations of arithmetic or logic operation
4. How a word can be fetched from memory?
5. What actions are necessary to execute bus instruction?
6. What are all the sequence of control steps required to perform these operations for the single bus architecture?
7. What is The control sequence for unconditional branch instruction ?

Explain hardwired control?(Nov

8. Explain about complete processor?
9. Explain about microprogrammed control?
10. What are all the components of microprogrammed control unit?
11. What are the Advantages of Microprogrammed control?(April 2015)
12. What are the Disadvantages of microprogrammed control unit?
13. Explain about Microinstruction.
14. What are the signals of microinstruction?
15. Give the difference between horizontal and vertical organization?

Give the Advantages of vertical and horizontal organization

16. Give the Disadvantages of vertical and horizontal organization
17. What are the 2 important factors must be considered while designing the microprogram sequencer?
18. What are 3 sources it has during execution of a microprogram the address of the next microinstruction to be executed?
19. Give the Comparison between Hardwired and Microprogrammed Control(Nov 2014)
20. Give the basic concepts of pipelining?
21. Define data hazard?(Nov 2015)
22. Explain about Addressing modes(Nov 2014,2015)
23. What are the Features of modern processor addressing modes.
24. What are The operations the processor can perform independently?
25. Define superscalar operations?

11 Marks

1. Give note on (i) Complete Instruction Execution (ii) Microprogrammed Control(Nov 2015)
2. What are data and instructional hazards and how this can be avoided?(Nov 2015)
3. Describe the techniques for handling data and instructional hazards in pipelining.(April 2015)

4. Explain the basic organization of a microprogrammed control unit and the generation of control signals using microprogram.(April 2015)
5. Explain the concept of hardwired control with neat diagram.(Nov 2014)
6. What is meant by hazard? Explain the types of hazard with examples.(Nov 2014)
7. Explain any two types of multiple processor organizations to speed up the processing.
8. Describe the techniques for handling data and instructions hazards in pipeline.(April 2015)
9. Explain in detail about superscalar operations.
10. Explain in detail about multiple bus organization.

Unit - I

1. Functional units of a computer
2. Addressing modes
3. Shift & Rotate instructions
4. Basic Instruction types.
5. Stacks & Queue

Unit - II

1. IA - 32 instructions
2. Various addressing modes available in IA32 pentium
3. Architecture of IA - 32 with block diagram
4. Subroutine ~~exp~~ concept in IA - 32 Architecture.
5. IA - 32 Assembly language.

Unit - III

1. DMA
2. Pentium Interrupt structure
3. Programmed I/O + M/y mapped I/O
4. Interrupts with diagram.

Unit - IV

1. Semiconductor RAM m/y
2. ROM
3. Cache M/y.
4. Virtual M/y
5. types of mapping functions in Cache m/y

Unit - V

1. Pipelining
2. Hazard
3. Instruction hazards
4. Micro programmed control unit
5. Various design methods of hardwired Control unit.